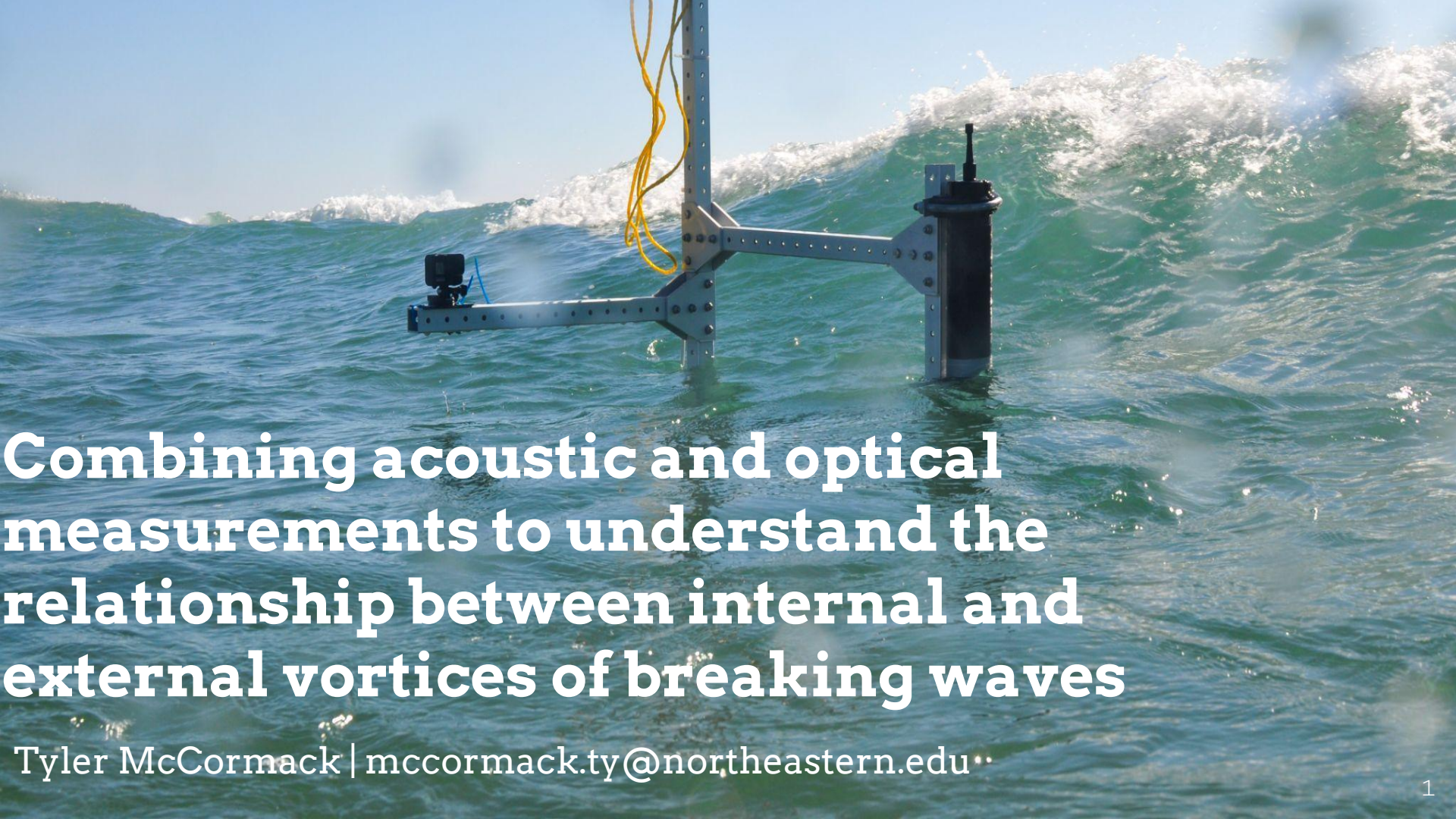


A scientific instrument, likely a combination of acoustic and optical sensors, is mounted on a metal frame in the ocean. The frame is partially submerged, with a vertical pole and horizontal arms. A yellow cable is visible hanging from the top. The background shows a large, breaking wave with white foam, indicating a dynamic ocean environment. The text is overlaid on the lower half of the image.

Combining acoustic and optical measurements to understand the relationship between internal and external vortices of breaking waves

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Presentation Outline

01 **Background**

+ + + + + +

02 **Research Question(s)**

+ + + + + +

03 **Methods**

+ + + + +

04 **Results**

+ + + + +

05 **Future Work**

+ + + + + +

+ + + + + +

Background

Sediment Transport

Particle Size, Accretion/ Recession



Nearshore Bathymetry

Bottom Contours, Slope, Water Depth

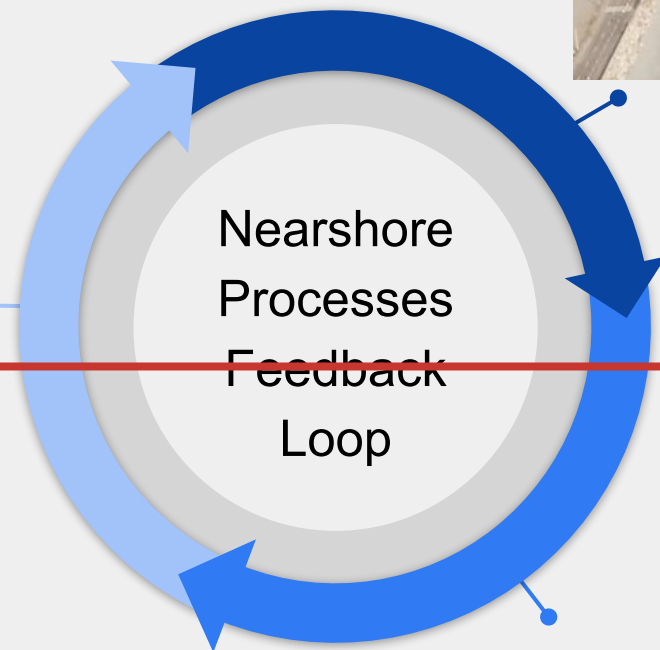
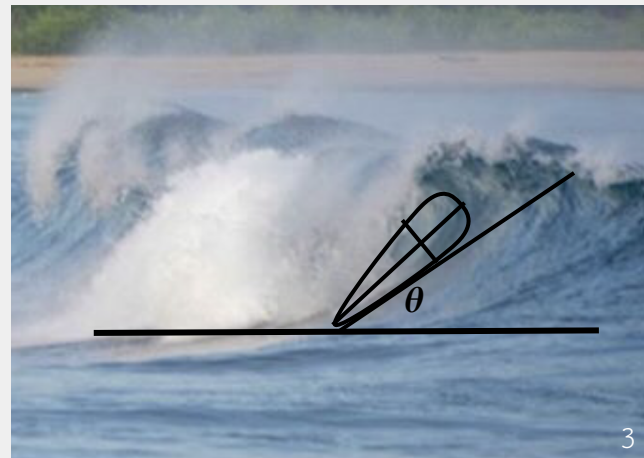


Figure 1. Nearshore Feedback Loop

Wave Characteristics

Wave Height, Wave Steepness, Vortex Shape, Overturning Jet Geometry



Research Question- Big Picture

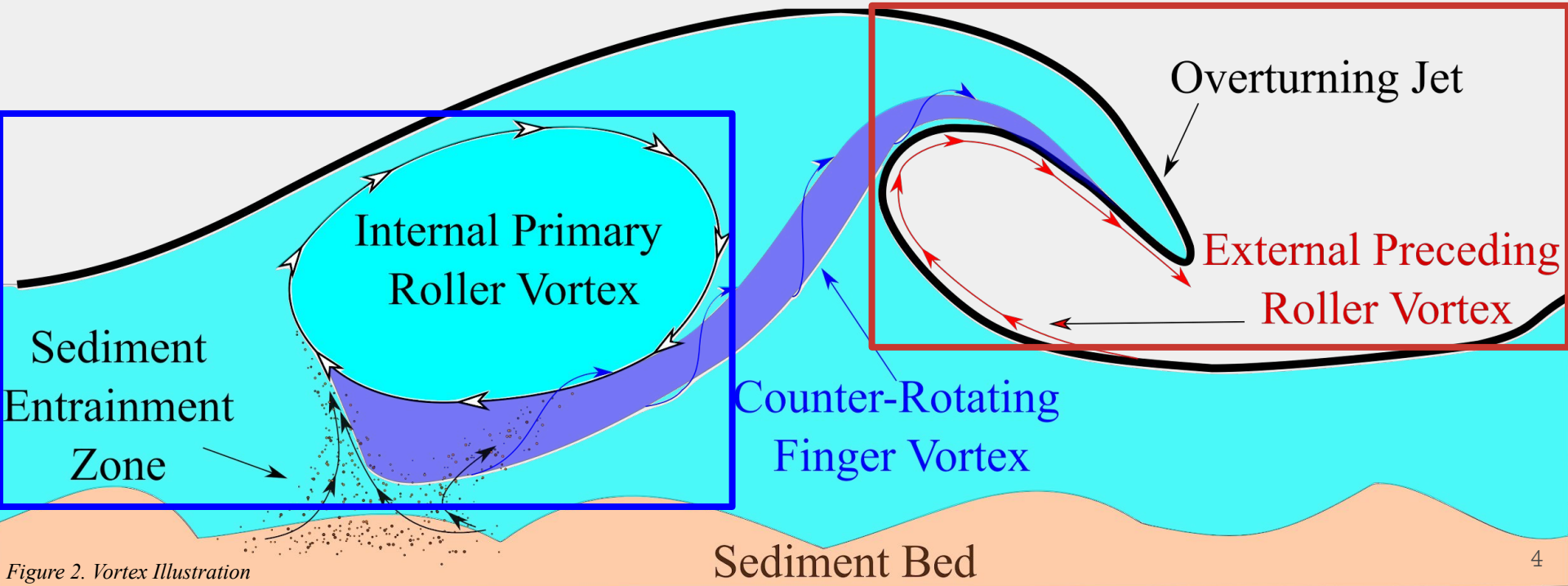
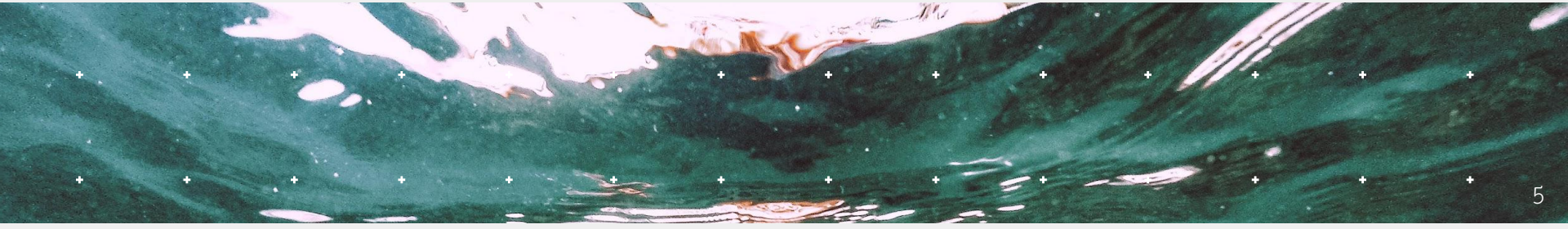


Figure 2. Vortex Illustration

Project Overview

- Hypothesis: External vortex geometry can act as a reliable indicator of expected internal vorticity
- Quantify properties of breaking waves' external vortical motion with optical data (i.e. video cameras!)
 - Geometry (e.g. vortex height, vortex width, angle, degree of curvature)
 - Dynamics (e.g. velocity, vorticity)
- Potential to alleviate the need of some in-situ instrument deployment



Project Overview

1

Assess the ability
of **close** range,
in-situ cameras to
accurately
capture external
vortical motion

2

Quantify
relationship
between external
and internal
vortices with in-situ
cameras and
current meters

3

Use **shorebased**
cameras and
in-situ current
meters to validate
relationship

Methodology

- Acoustic Doppler Velocimeter (ADV):
Point velocity measurements of
internal orbital velocities
 - Nortek Vector
 - Sampling Frequency: 64Hz
- Externally and internally mounted
cameras
 - GoPro Hero9
 - Sampling Frequency/ Frame rate:
120 frames per second (FPS)

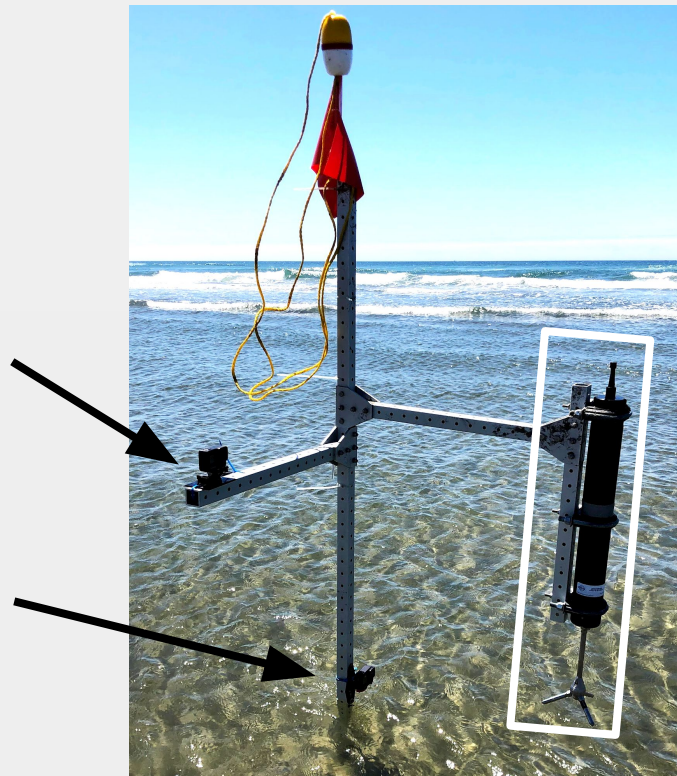
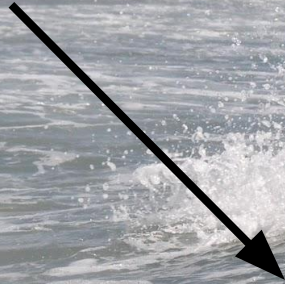


Figure 3. Instrumentation Diagram

Methodology

**Externally
mounted camera**



**Acoustic Doppler
Velocimeter (ADV)**



Methodology

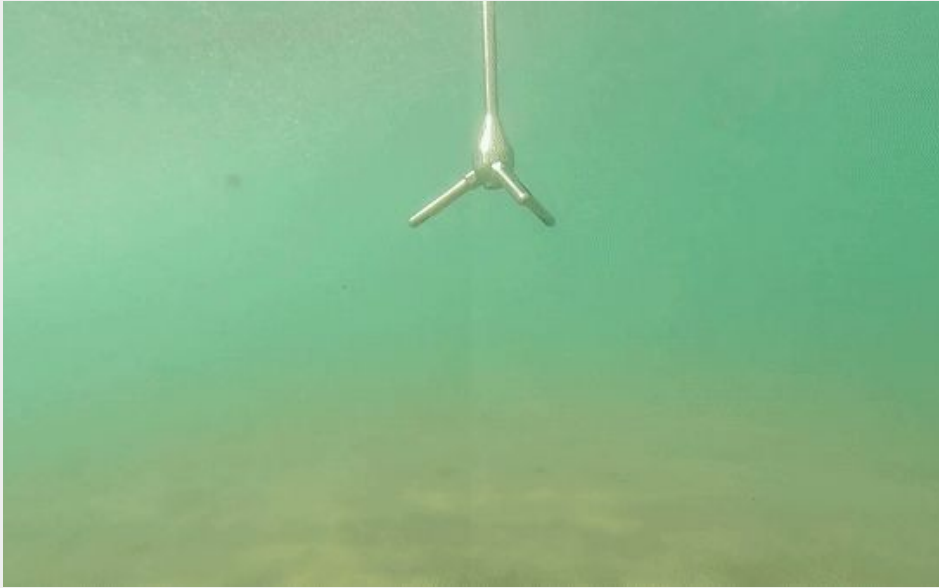


*Figure 5a. Externally mounted camera view,
Longshore facing*



Figure 5b. Externally mounted camera callout

Methodology



*Figure 6a. Internally mounted camera view,
Longshore facing*



Figure 6b. Externally mounted camera callout

Methodology

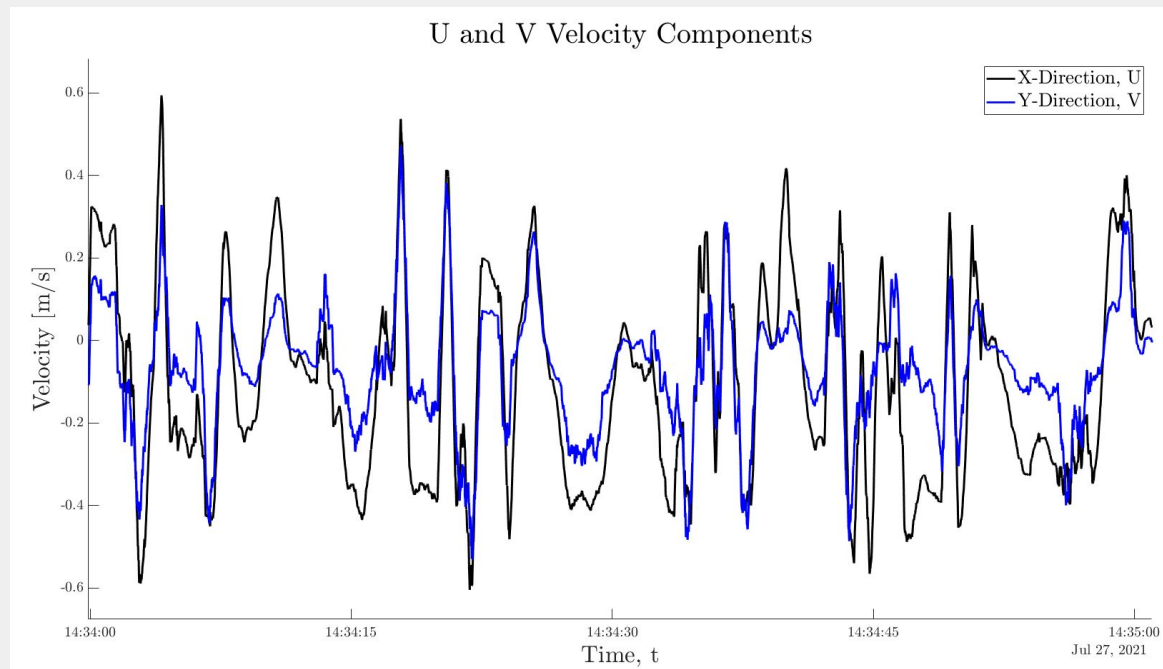


Figure 7a. Example ADV U and V velocities [m/s] data



Figure 7b. ADV callout

Data Analysis

- Particle Image Velocimetry (PIV)
 - Measures displacements between frames [pixels]
 - Apply a calibration with a known distance for pixel to meter ratio
 - Calculates velocity with measured distance and time between frames
 - Camera frame rate
- Bubble Image Velocimetry (BIV)

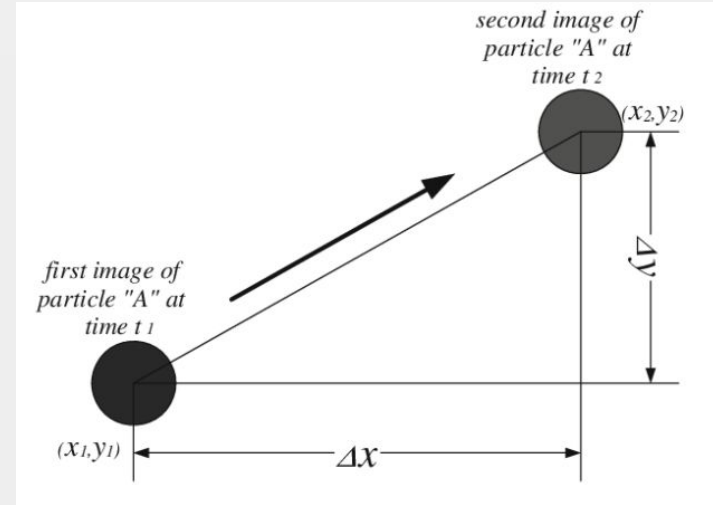
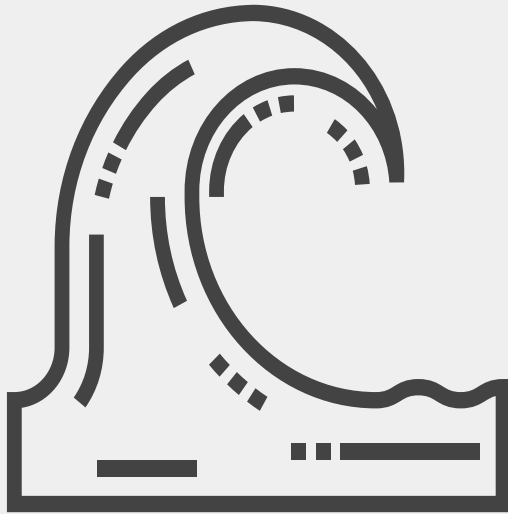


Figure 8. PIV operating principle diagram (Lee et al., 2010)

Preliminary Results



Internal Results

- Extensive post processing required
- Depth poses significant issues
 - Bubble inundation
- Capable of resolving 3D vortical motion
 - Can not quantify third dimension



Internal Results

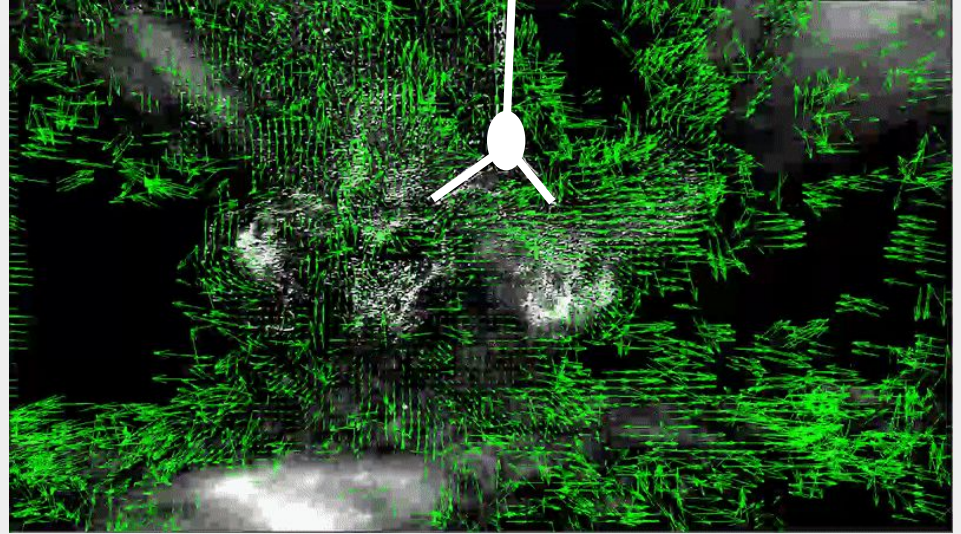


Figure 9. PIV vectorization for an internal vortex beneath a breaking wave



External Results

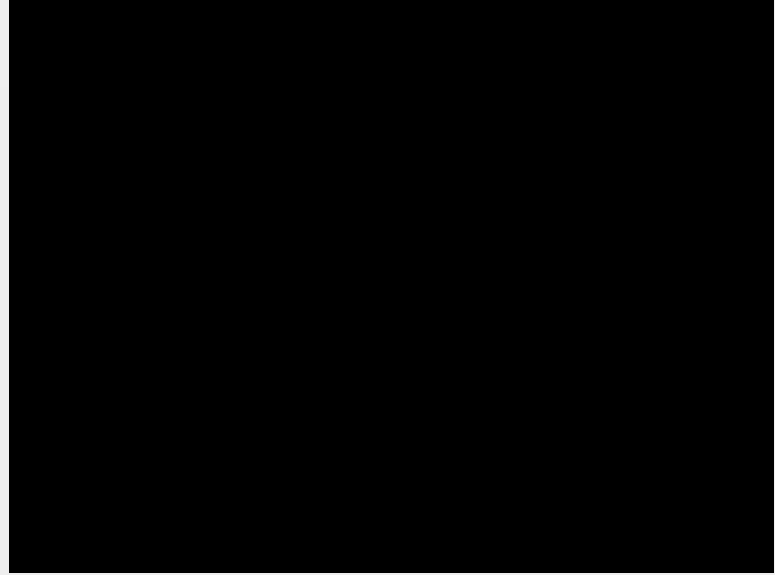


Figure 10. PIV vectorization of wave run up from a downwards facing camera



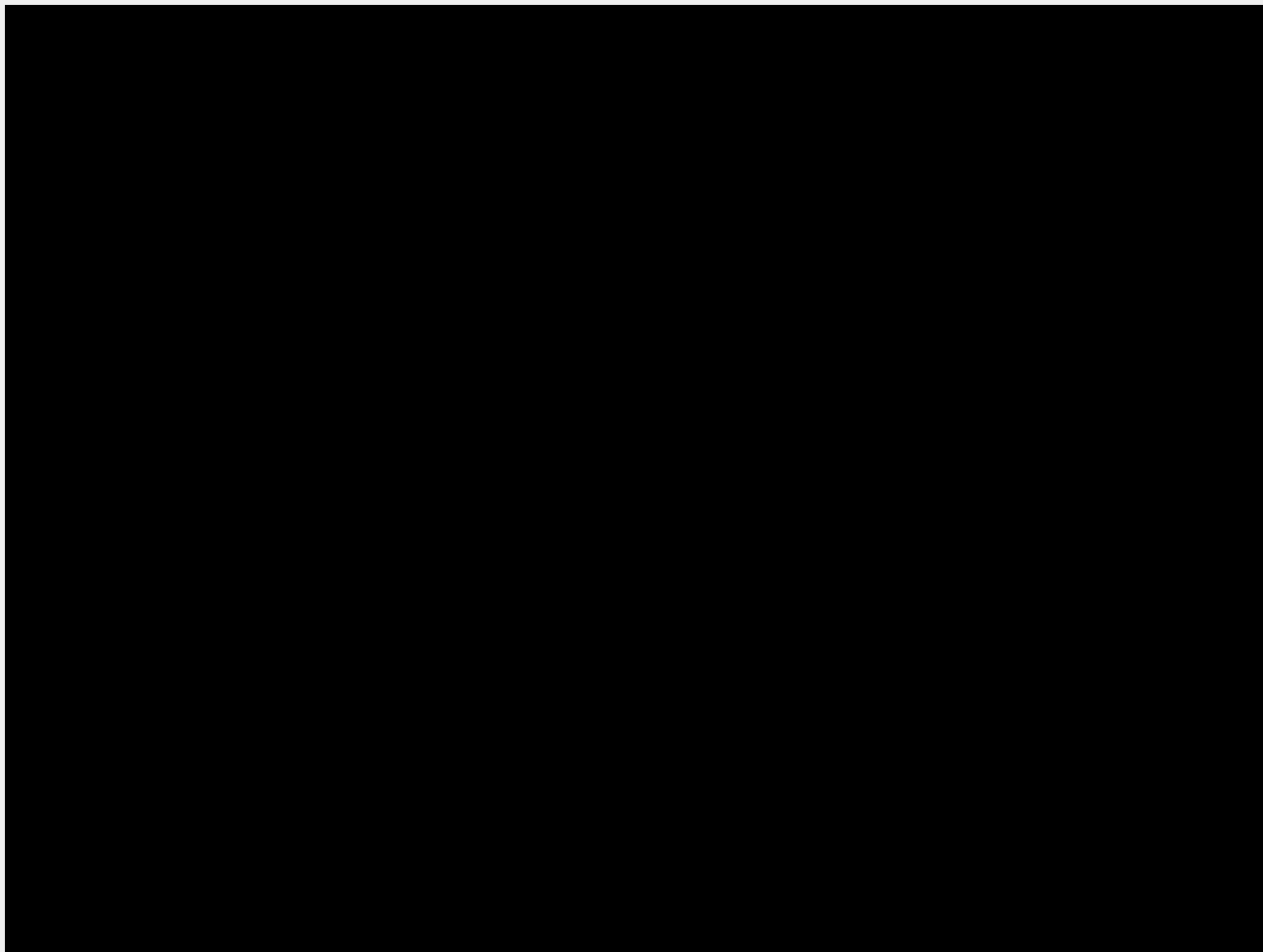


Figure 11a. Downward facing camera (top) b. Corresponding PIV derived UV velocity magnitude (bottom)

External Results

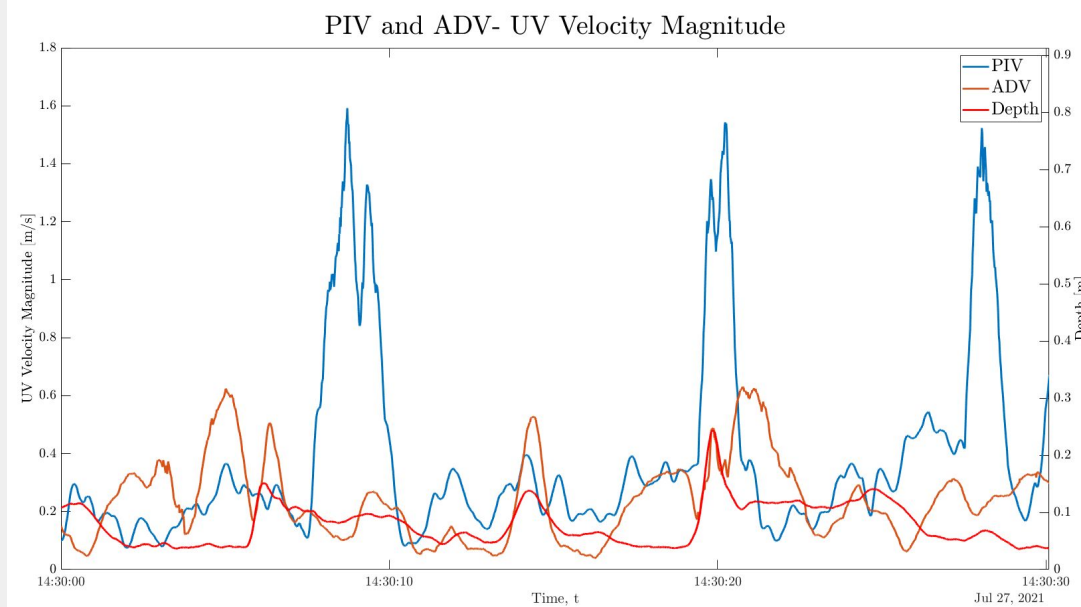


Figure 12. UV velocity magnitude measured by PIV (blue) and ADV (orange), water depth (red)

- Correct order of magnitude
- Apparent velocity spike due to camera submersion
- Unidentified phase lag due to clock drift in camera



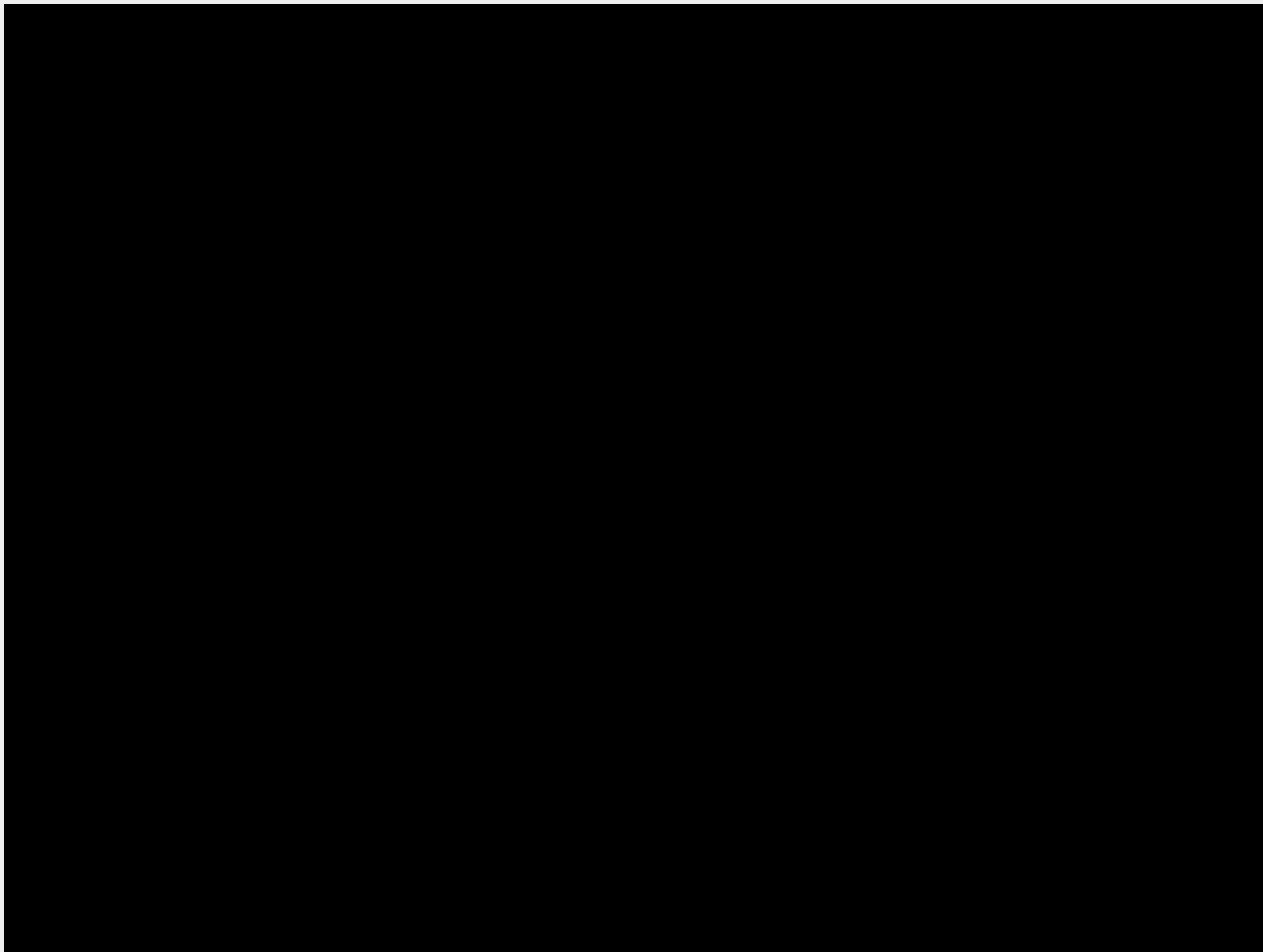


Figure 13a. Longshore facing camera (top) b. Corresponding PIV derived XY velocity magnitude (bottom)

Conclusions

- Consumer grade video cameras and particle image velocimetry are capable of characterizing the external vortex of a breaking wave
- This combination of instruments has the ability to capture and quantify the internal and external vortical motion relationship
- Further analysis required to quantify velocity deviations
- Extremely inconsistent clock in GoPro Hero9
- Depth of field poses challenges, particularly to the internal camera

Next Steps

- Determine the range of conditions where PIV provides acceptable results
- Test stereoscopic camera systems
- Gather a close range video dataset of breaking waves
- Gather shorebased or aerial dataset of breaking waves with an in-situ ADV

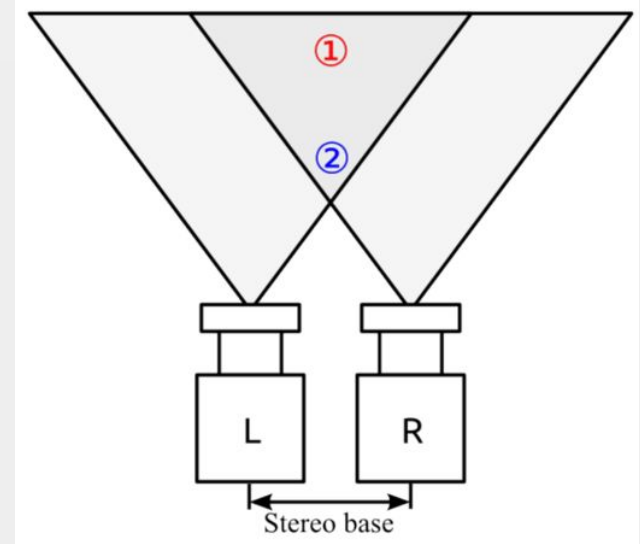


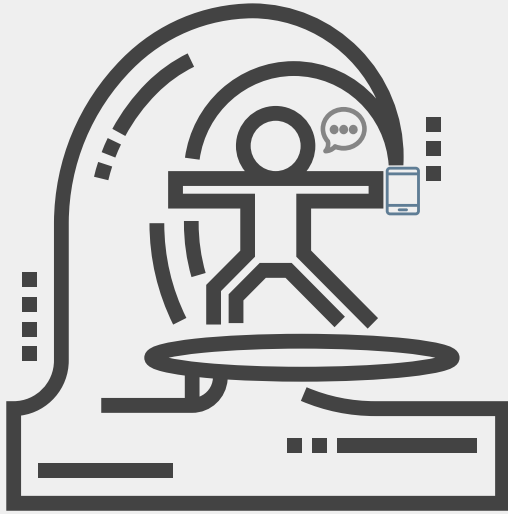
Figure 14. Stereoscopic Camera Diagram



References

Lee, Kyoungsoon, et al. "Measurement of Swimming Speed of Giant Jellyfish *Nemopilema Nomurai* Using Acoustics and Visualization Analysis." *Fisheries Science*, vol. 76, Nov. 2010, pp. 893–99. *ResearchGate*, <https://doi.org/10.1007/s12562-010-0294-7>.

Get it Touch!



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